

La ROSalía

Puzzlebot Autonomous exploration

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EKF Localisation

Prediction Model

The prediction step estimates state transitions by integrating high-frequency wheel odometry kinematic calculations.

Correction Model

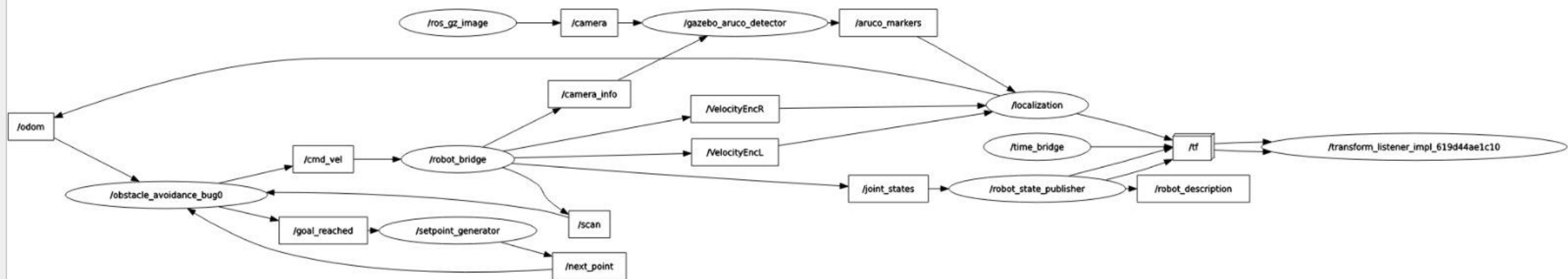
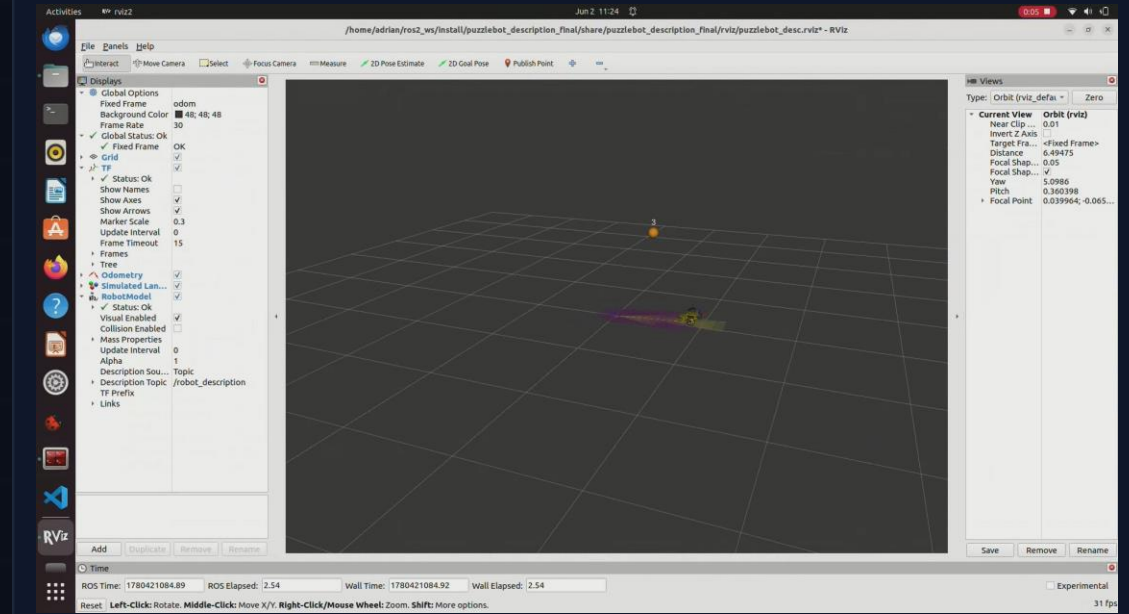
Applies observations of artificial **ArUco Markers** to adjust accumulated localization drift.

Algorithm: EKF Localisation with known data association

1. **function** EKF-Localisation ($\mathcal{M}, \mu_{k-1}, \Sigma_{k-1}, \mathbf{u}_k, \mathbf{z}_{i,k}, \mathbf{Q}_k, \mathbf{R}_k$)
2. $\hat{\mu}_k \leftarrow \mathbf{h}(\mu_{k-1}, \mathbf{u}_k)$
3. $\mathbf{H}_k \leftarrow \nabla_{s_{k-1}} \mathbf{h}(s_{k-1}, \mathbf{u}_k)|_{s_{k-1}=\mu_{k-1}}$
4. $\hat{\Sigma}_k \leftarrow \mathbf{H}_k \Sigma_{k-1} \mathbf{H}_k^T + \mathbf{Q}_k$
5. **if** $\mathbf{z}_{i,k}$ corresponds to landmark $\mathbf{m}_i \in \mathcal{M}$
6. $\hat{\mathbf{z}}_{i,k} \leftarrow \mathbf{g}(\mathbf{m}_i, \hat{\mu}_k)$
7. $\mathbf{G}_k \leftarrow \nabla_{s_k} \mathbf{g}(\mathbf{m}_i, s_k)|_{s_k=\hat{\mu}_k}$
8. $\mathbf{Z}_k \leftarrow \mathbf{G}_k \hat{\Sigma}_k \mathbf{G}_k^T + \mathbf{R}_k$
9. $\mathbf{K}_k \leftarrow \hat{\Sigma}_k \mathbf{G}_k^T \mathbf{Z}_k^{-1}$
10. $\mu_k \leftarrow \hat{\mu}_k + \mathbf{K}_k (\mathbf{z}_{i,k} - \hat{\mathbf{z}}_{i,k})$
11. $\Sigma_k \leftarrow (\mathbf{I} - \mathbf{K}_k \mathbf{G}_k) \hat{\Sigma}_k$
12. **return** μ_k, Σ_k

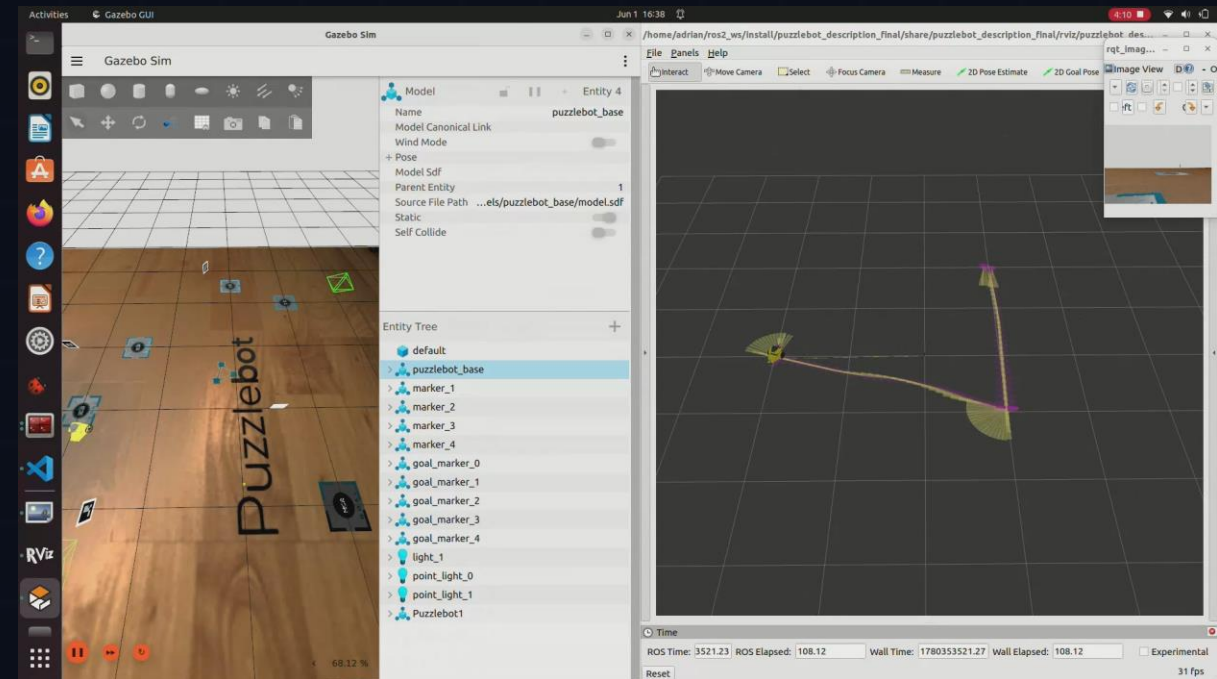
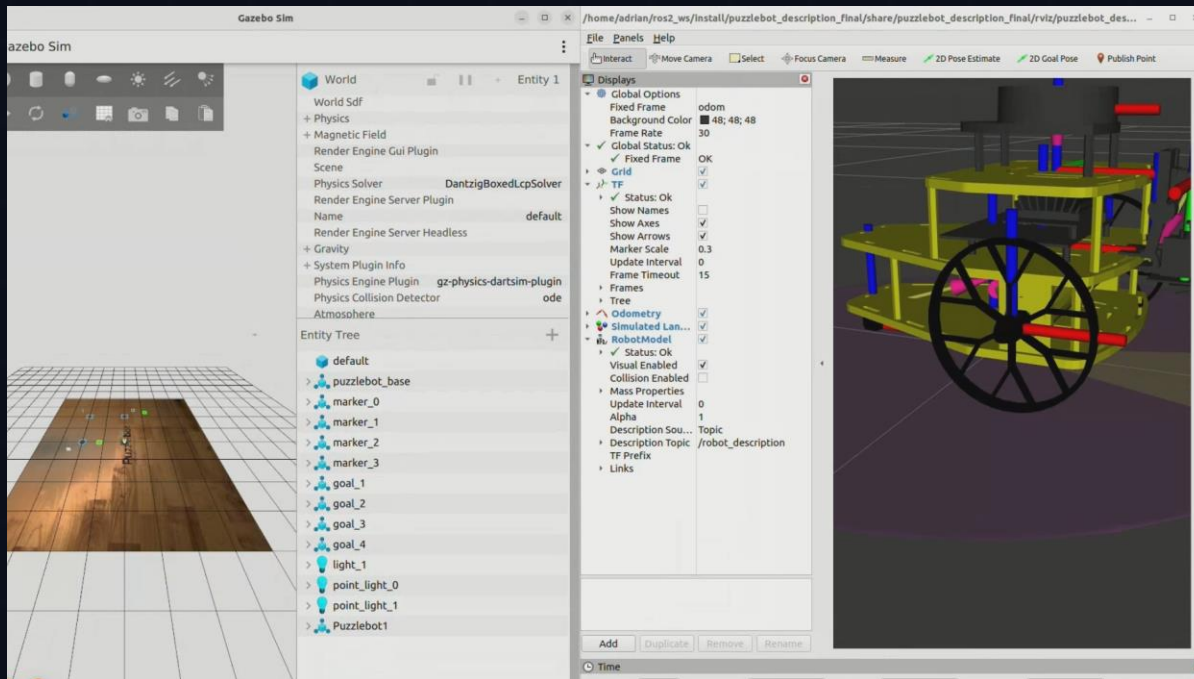
RVIZ & Algorithm Diagram

-  Covariance Growth
-  Ellipsoid Plotting
-  Robustness Validation



Simulation & Launch

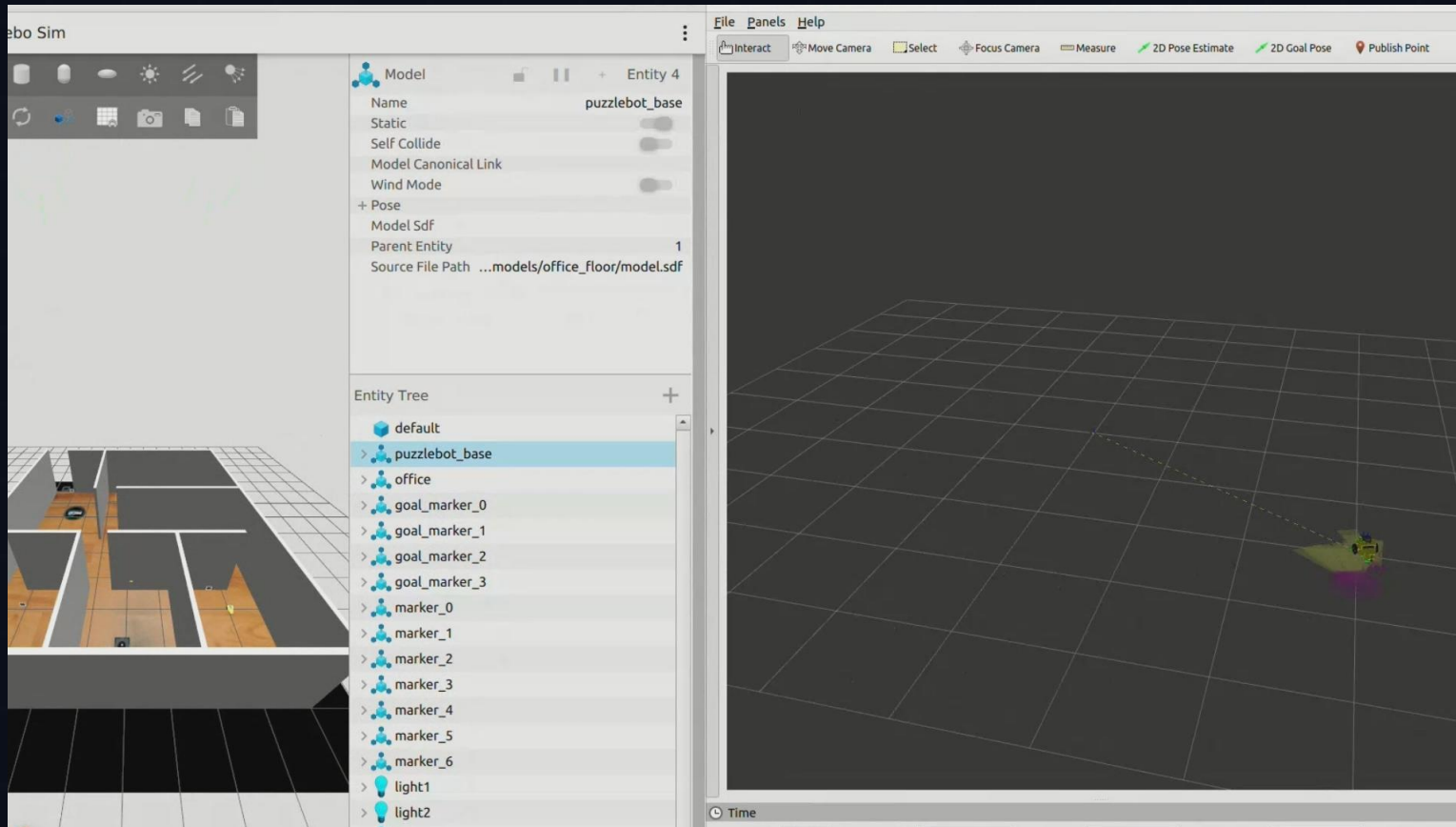
-  **Gazebo Environment:** Simulating active mapping environments under realistic physics constraints.
-  **Workflow Diagram:** Visualization of ROS architecture, communication channels, and topics.
-  **Launch Files:** Orchestrating all vision, estimation, and control nodes seamlessly.



Final Route

- 🛡️ **Collision Avoidance:** Seamless integration of distance sensors with path planner.
- 🔄 **Real-time Replanning:** Immediate alternate trajectory generation during exploration.
- ⚙️ **Uncertainty Management:** Actively managing position estimates while bypassing objects.

VIDEO PRESENTATION: <https://www.youtube.com/watch?v=UIXBX66iEXM>



Challenges faced



Covariance Growth

Preventing uncertainty explosion during long intervals of blind travel without any visible ArUco reference markers in sight.



Sampling Constraints

Optimizing simulation sampling time across concurrent ROS vision algorithms to prevent controller lag and physical overshoot.